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LAB SESSION 2: REQUIEMENT ANALYSIS

LIBRARY MANAGEMENT SYSTEM

Functional Requirements:

1. The system shall allow admin to create and block accounts of students as well as librarian.
2. The system shall allow user to search for books by title, author, Id, language, etc.
3. The system shall store the data of each student's borrow and return history and generate analysis report at the end of the month.
4. The system should show "not available" statement if the book is not available or borrowed by someone else.
5. The system shall allow librarian to add or remove books.
6. The system shall allow the user to pre-book/reserve the book that is currently unavailable.
7. The system shall send a notification to user when the reserved book becomes available.
8. The system shall allow user to cancel reservation.
9. The system shall charge fine to users if they did not return book after due date: Rs. 100 per day.
10. The system shall send a notification to user 2 days before the due date.

Non-Functional Requirements:

1. Responsiveness: The UI should work in PCs as well as mobile phones.
2. Secure: The system should be secured from viruses and hackers.
3. Availability: The system should be always available when user tries to use it.
4. Scalability: The database must support large number of book records.
5. Fast: The system should be fast enough to response within 2 seconds.
6. Reliability: The system shall ensure data consistency during all book transactions.

7. Backup and Recovery: The system should automatically backup the database every 24 hours.
8. Portability: The system should run on Windows, Linux and macOS platforms.
9. Maintainability: The system shall support easy updates without affecting existing functionalities.
10. Resource Efficiency: The system shall support high number of concurrent users without significant performance degradation.